

Junzhao Yang

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EDUCATION

Carnegie Mellon University, Computer Science Department 2024–present
Ph.D. in Computer Science (expected 2027). Advisor: Richard Peng
Research areas: numerical linear algebra, spectral graph theory

Tsinghua University, Institute for Interdisciplinary Information Sciences 2020–2024
B.S. in Computer Science, Yao Class.

PROJECTS

FizzyProver May 2026–present
Sole developer · Multi-agent system for mathematical research

- Designed a recursive Codex workflow that schedules independent proof subproblems in parallel and alternates prover and verifier passes to critique and revise proof attempts.
- Built and deployed a web platform used by 20+ academic users across 300+ runs. Among the runs we have examined to date, including runs powered by GPT-5.5, we have not observed any incorrect final proofs; by contrast, our direct GPT-5.5 Pro attempts produced flawed proofs.

HONORS AND AWARDS

- ICPC North America Championship 2025, 2nd place, Silver Medal.
- **46th ICPC World Finals, 10th place, Bronze Medal** (held in 2024).
- **International Olympiad in Informatics 2019, 5th place, Gold Medal.**
- National Olympiad in Informatics 2018, Gold Medal.
- Yao Award, Bronze Medal (awarded to five Yao Class students in 2023).

PUBLICATIONS

Authors are listed alphabetically, as is customary in theoretical computer science.

- [1] **Fast Convergence of Gradient Descent on Random ℓ_p -Norm Regression** **FOCS 2026 (to appear)**
Yang Liu, Richard Peng, Weina Wang, **Junzhao Yang**
- [2] **Approximate Spanning Tree Counting from Uncorrelated Edge Sets** **FOCS 2026 (to appear)**
Yang P. Liu, Richard Peng, **Junzhao Yang**
- [3] **Entrywise Approximate Solutions for SDDM Systems in Almost-Linear Time** **STOC 2026**
Angelo Farfan, Mehrdad Ghadiri, **Junzhao Yang**
- [4] **Entrywise Approximation for Matrix Inversion and Linear Systems** **SODA 2026**
Mehrdad Ghadiri, Hoai-An Nguyen, **Junzhao Yang**
- [5] **Numerical Linear Algebra in Linear Space** **SODA 2026**
Yiping Liu, Hoai-An Nguyen, **Junzhao Yang**
- [6] **Memory-Sample Lower Bounds for LWE** **CRYPTO 2024**
Mingqi Lu, **Junzhao Yang**
- [7] **The Cost of Parallelizing Boosting** **SODA 2024**
Xin Lyu, Hongxun Wu, **Junzhao Yang**
- [8] **Tight Time-Space Lower Bounds for Constant-Pass Learning** **FOCS 2023**
Xin Lyu, Avishay Tal, Hongxun Wu, **Junzhao Yang**